

**The Obesity Epidemic:
A Novel Analysis of Patient-Level Obesity and Machine
Learning Risk Mitigation Optimization**

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Contents

1	Executive Summary	4
2	Background Information	5
2.1	Defining Obesity	5
2.2	Importance of Project	5
2.3	Target Audience	5
3	Data Methodology	5
3.1	UCI Machine Learning Repository	6
3.1.1	Data Reliability	6
3.2	National Health and Nutrition Examination Survey	7
4	Mathematics Methodology and Analysis	7
4.1	Model Overview	7
4.2	Phase 1: Patient-Level Final Classification Models	8
4.2.1	XGBoost Architecture	8
4.2.2	Optimized XGBoost Methodology	9
4.2.3	XGBoost Performance Metrics	10
4.2.4	FTTransformer Architecture and Methodology	13
4.3	Feature Tokenization and Transformer Blocks	13
4.3.1	FTTransformer Results	15
4.4	Logistic Classification	17
4.4.1	Logistic Classification Architecture	17
4.5	Model Comparison	17
4.6	Phase 3: Per-Sample Monte Carlo Risk Mitigation Optimization	18
4.6.1	Part 1: Obesity	18
4.6.2	Part 2: Overweight	20
4.7	Markov Chain: Risk Mitigation Impact Modeling	21
4.7.1	Baseline Transition Matrix Estimation	21
4.7.2	Intervention-Adjusted Transition Model	21
5	Risk Analysis	22
5.1	Defining the Risks	22
5.1.1	Personal Health Risks	22
5.1.2	Societal and Economic Risks	22
5.1.3	Sensitivity Analysis	23
5.1.4	Groups at Risk	24
5.1.5	Severity	24
5.1.6	Likelihood	24
5.1.7	Uncertainty	24
5.1.8	Expected Value	25
6	Recommendations	25
6.1	Behavior Change	25
6.2	Modifying Outcomes	25
6.2.1	Personalized Lifestyle Adjustments	25
6.3	Insurance	25
6.3.1	Overview	25
6.3.2	Incentives and Rewards Structure	25
6.3.3	Premium Structure	26
6.3.4	Coverage for Obesity-Related Diseases	26
6.3.5	Implementation and Monitoring	26

6.3.6	Markov Chain Results and Long-Term Impact	26
7	Conclusion	28
8	Bibliography	29

1 Executive Summary

Obesity presents a major global health challenge, increasing the risk of cardiovascular diseases, diabetes, and other chronic conditions. Additionally, the economic impact is substantial, with obesity-related healthcare costs exceeding \$173 billion annually in the U.S. alone (1). This study develops a machine learning-driven framework for assessing obesity risk and generating personalized lifestyle recommendations. By classifying individuals into seven weight categories using XGBoost and FTTransformer, we leverage Monte Carlo simulations to create targeted interventions aimed at reducing obesity probability while mitigating underweight risk.

Using the UCI Machine Learning Repository dataset (2,111 samples, 17 features), we trained three classification models: (2)

- **XGBoost**: Achieved 85.82% accuracy, demonstrating strong performance in distinguishing severe obesity cases.
- **FTTransformer**: Attained 82.00% accuracy, capturing complex feature interactions but slightly underperforming compared to XGBoost.
- **Multinomial Logistic Classification (Baseline)**: Performed significantly worse at 53.00% accuracy, serving as a benchmark for assessing improvements.

XGBoost emerged as the most effective model, particularly in distinguishing severe obesity cases, making it the chosen model for intervention optimization.

To generate effective lifestyle modifications, we implemented a Monte Carlo-based recommendation framework. The model adjusted four key lifestyle factors:

- **Physical Activity Frequency (FAF)**: Increased activity levels to promote energy expenditure and weight management.
- **Calorie Consumption Monitoring (SCC)**: Encouraged daily tracking of caloric intake to improve dietary awareness.
- **Consumption of Food Between Meals (CAEC)**: Recommended reducing snacking frequency to prevent excessive calorie intake.
- **Daily Water Consumption (CH₂O)**: Increased hydration to support metabolic regulation and overall health.

For each individual, 500 candidate interventions were generated, and the optimal modification was selected based on an objective function that minimized obesity probability while preventing excessive underweight risk. As a result, obesity probability was reduced by 36.14%, while overweight probability decreased by 13.5%, underscoring the effectiveness of data-driven interventions.

A sensitivity analysis revealed that calorie monitoring and physical activity frequency had the greatest impact on classification outcomes, suggesting that structured behavioral modifications can meaningfully reduce obesity risk.

We designed a behavior change risk mitigation plan in order to reduce the probability of becoming overweight or obese. This study successfully integrates machine learning classification with Monte Carlo-based intervention optimization, providing a scalable, personalized framework for obesity risk mitigation. We developed modifying-outcomes recommendations for obese and overweight individuals with objective numerical evidence and statistical backing of their robustness. We then developed an insurance plan and incentives to include these recommendations as part of the plan. We evaluated the long-term effects of our recommendations with a discrete-time Markov Chain using a transition matrix estimated from the National Health and Nutritional Survey dataset for a hypothetical insurance company with 100,000 policyholders that are representative of the US weight class distribution (3).

2 Background Information

2.1 Defining Obesity

Obesity is a chronic health condition characterized by having an excessive amount of body fat. To determine if someone is obese, the body mass index (BMI) is calculated by dividing the person's weight by the square of the person's height; the numerical value of the BMI is then compared to standard BMI categories to determine obesity status.

Table 1 details how different BMIs are categorized by the Centers for Disease Control (4):

CDC BMI Category	Range
Insufficient Weight	< 18.5
Normal Weight	$18.5 - 24.9$
Overweight Level	$25.0 - 29.9$
Obesity Type I	$30.0 - 34.9$
Obesity Type II	$35.0 - 39.9$
Obesity Type III	≥ 40.0

Anyone with a BMI of 30 or above is considered obese. Obesity is further categorized into three classes of increasing severity: Class I (BMI 30–34.9), Class II (BMI 35–39.9), and Class III (BMI 40 or higher) (5).

2.2 Importance of Project

Obesity is increasingly prevalent in modern-day society, especially in the US America's obese population has been growing steadily over the past century, starting at 12% in 1990 and more than tripling by 2023 (6; 7; 4). From 2021 to 2023, around 40.3% of American adults were obese (4).

Obesity puts the individual at risk for other potentially life-threatening conditions such as increased risk for high blood pressure, high cholesterol, and type 2 diabetes—all of which contribute to a heightened likelihood of heart disease. Adults dealing with obesity also face an increased likelihood of experiencing a range of cancers and mental health disorders such as depression and anxiety. Moreover, obesity is associated with respiratory challenges like asthma and sleep apnea, as well as joint issues including osteoarthritis and other conditions such as gallstones and gallbladder disease (8).

Obesity also poses significant negative consequences at the societal level. Direct medical costs for prevention, diagnosis, and treatment cost the American healthcare system nearly \$173 billion every year, and the cost is compounded by indirect costs such as absenteeism and reduced professional performance that lead to an additional loss of \$6.38 billion (8).

Obesity can be caused by a variety of factors, such as lack of physical exercise, unhealthy diets and eating patterns (such as frequent consumption of heavily processed foods), poor sleep, and genetic makeup (9). Many of these factors causing obesity are preventable and reversible (10). Essentially, obesity is caused by an excessive surplus of calorie consumption and inadequate calorie expenditure.

2.3 Target Audience

Treatments to prevent and manage obesity work by reducing calorie intake, increasing calorie expenditure, or often a combination of both. Our model aims to generate optimal modifying outcome recommendations on physical activity, snacking habits, calorie tracking, and water consumption that are unique for each individual's weight class.

3 Data Methodology

We used the “Estimation of Obesity Levels Based On Eating Habits and Physical Condition” from the University of California Irvine Machine Learning Repository dataset in order to develop a classification model for patient-level obesity using tabular data. We also used the National Health and Nutritional Exam Survey dataset to create the Yearly Markov Transition Matrix to simulate the long-term impacts of our recommendations (2; 3).

3.1 UCI Machine Learning Repository

The UCI Machine Learning Repository is a dataset that allows us to characterize/refine categories of potential outcomes. This dataset is composed of 17 features and 2,111 samples.

Feature	Description	Cardinality
Gender	Participant’s gender	2
Age	Age in years	Numeric
Height	Height in meters	Numeric
Weight	Weight in kg	Numeric
Family History	Overweight history (Yes/No)	2
FAVC	High-calorie food intake (Yes/No)	2
FCVC	Vegetable consumption frequency	3
NCP	Daily main meal count	3
CAEC	Snacking frequency	4
SMOKE	Smoking status (Yes/No)	2
CH2O	Daily water intake	3
SCC	Calorie monitoring (Yes/No)	2
FAF	Physical activity frequency	4
TUE	Daily tech device usage	3
CALC	Alcohol consumption frequency	4
MTRANS	Primary transportation mode	5
NObesyesdad	Obesity classification	7

Table 1: Obesity Dataset Features and Cardinalities

BMI Category	Range
Insufficient Weight	< 18.5
Normal Weight	18.5 – 24.9
Overweight Level I	25.0 – 27.4
Overweight Level II	27.5 – 29.9
Obesity Type I	30.0 – 34.9
Obesity Type II	35.0 – 39.9
Obesity Type III	≥ 40.0

Table 2: NObesyesdad BMI Categories

Our target variable, *NObesyesdad*, is a categorical feature with seven weight classes: Insufficient Weight, Normal Weight, Overweight Level I, Overweight Level II, Obesity Type I, Obesity Type II, and Obesity Type III.

Note: The dataset’s obesity types differ from NIH obesity classes. Type I and II correspond to lower and upper halves of NIH Class II obesity (BMI 30.0–39.9 kg/m²), while Type III aligns with NIH Class III obesity (≥ 40 kg/m²).

Preprocessing: We apply label encoding to categorical variables, as XGBoost benefits from integer-encoded features without additional scaling. Continuous features undergo Z-score normalization (mean = 0, std = 1) to balance feature contributions. We dropped the Weight column, which is a major aspect of the BMI calculation, which is used to define the 7 classes.

There is a major imbalance in Obesity Type II and III across genders—females are extremely underrepresented in Type II but overrepresented in Type III, with the inverse for males, so we dropped the Gender column to prevent bias from this column.

Dataset Splitting: The dataset is divided into training, validation, and testing sets. Given the small validation/testing sizes, K-fold cross-validation ensures robust evaluation.

3.1.1 Data Reliability

77% of this dataset was generated synthetically using Weka and SMOTE filters, with 23% of the data obtained directly from users through a web platform in the countries of Mexico, Peru, and Columbia (2). The dataset is completely devoid of NaNs and has near symmetrical class distribution. The dataset consists of a relatively small sample of 2,111 rows, so larger and more complex models may struggle to generalize.

The highest class, Obesity Type I, has around 350 samples. However, the other classes are not far behind in count, with the lowest being Insufficient Weight, which has 272 samples.

3.2 National Health and Nutrition Examination Survey

The National Health and Nutrition Examination Survey (NHANES) spans 1988 to 2018, covering questionnaires, lab reports, and demographics (3). We selected specific years, weight-class identifiers (Normal Weight, total Overweight and Obesity, total Obesity, and three Obesity types), and distribution estimation features.

The NHANES dataset allows us to define severity or range of loss of potential outcomes, and define frequency or likelihood of potential outcomes. It includes both information on weight class and weight severity as well as the yearly frequency of such classes.

4 Mathematics Methodology and Analysis

4.1 Model Overview

Our model methodology includes both patient-level modeling for generating optimal recommendations and national-level modeling in order to determine the long-term frequency and impact of obesity. Our goal is to use machine learning and algorithmic approaches to mathematically generate the optimal Modifying Outcomes recommendations for different weight classes (Overweight and Obese). We separated this procedure into three phases:

During Phase 1, we develop an XGBoost model and a Feature-Tokenzer Transformer, with hyperparameters fine-tuned with the Tree-structured Parzen Estimator surrogate model (11). We then compare the results along with a baseline Logistic Classification model, and select the best model to move on to Phase 3 of our Mathematics Methodology.

Phase 2 is two Per-Sample Monte Carlo Risk Mitigation Optimization using the previous patient-level model as the forward model to give optimized and unique recommendations for Overweight individuals as well as Obese individuals. This will output a range of unit changes for Physical Activity (FAF), Calorie Consumption Monitoring (SCC), Consumption of Food Between Meals (CAEC), and Water Consumption (CH2O) as well as the estimated reduction in obesity / overweight probability. We further evaluate this phase with Kernel Density Estimation (KDE) of the probability distributions before applying our recommendations and after applying our recommendations. Our recommendations are modifying outcomes recommendations to help obese or overweight individuals lose weight

Phase 3 is where we apply our modifying outcomes and behavioral change recommendations across a 20-year time span using a Markov Chain.

1. **Assumption:** The CDC definition of the Overweight class is having a BMI from 25 to 29.9. This class is split into two levels (Overweight Level I and II) in our dataset. We assume that Overweight Level I is the 25 - 27.4 BMI range, while Overweight Level II is the 27.5 - 29.9 BMI range.

Justification: In order to maintain the integrity of the dataset class distribution, we maintained the two-level definition of the Overweight class. The original paper that made this dataset defined the other classes following the CDC BMI ranges, so it can be assumed that Overweight Level I and Level II represent the first half of and second half respectively of the Overweight BMI range (2).

2. **Assumption:** The Markov Transition Matrices stay consistent each year.

Justification: This assumption ensures computability.

3. **Assumption:** In the NHANES dataset, because it was missing the insufficient weight column, we assumed that it has a negative linear relationship where it starts at the proportions of 2.3% in 1988 to 1.6% in 2018 (3; 12)

Justification: This assumption ensures cross-dataset computability.

4. **Assumption:** The NHANES dataset carries over to 2025.

Justification: Ensures computability by allowing us to use decades of historical trends to estimate obesity probabilities in the future.

5. **Assumption:** The number of insurance holders stays constant.

Justification: This assumption ensures computability.

6. **Assumption:** The the synthetically generated data from the UCI Machine Learning Repository accurately represents real-world obesity patterns and is reliable for training models.

Justification: This assumption ensures computability.

4.2 Phase 1: Patient-Level Final Classification Models

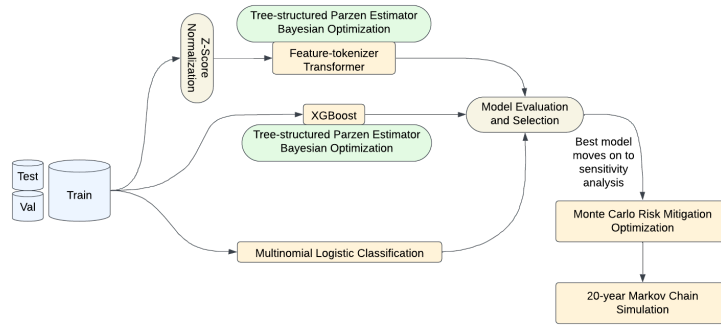


Figure 1: Phases 1 and 2 Overview

We developed XGBoost and Feature-Tokenizer Transformer models for multiclass classification of weight types for tabular input data (13). For both models, we apply Tree-structured Parzen Estimator surrogate models for Bayesian Optimization of hyperparameters to ensure optimal results (11). We evaluated each model with a variety of exhaustive metrics and compared them with a baseline Multinomial Logistic Classification model. The model with the best testing metrics moves on to Phase 3, where we use it as the forward model for the Per-Sample Monte Carlo Simulation.

4.2.1 XGBoost Architecture

Xtreme Gradient Boosting is a tree-boosting algorithm optimized for efficiency and strong generalization on tabular datasets, particularly for classification tasks (14).

Unlike individual decision trees, which are prone to underfitting, XGBoost builds multiple trees using sub-sampled rows and columns, enhancing predictive performance. In multiclass classification, it constructs separate trees for each class, summing their outputs before applying a softmax function to aggregate predictions. As a CART-based model, XGBoost predicts target values by accumulating the outputs of its trees.

Variable	Description
α	Lasso regularization parameter (L1 regularization)
λ	Ridge regularization parameter (L2 regularization)
γ	Minimum loss reduction required to make a further partition on a leaf node
Similarity Score	A measure of node purity, computed as the sum of squared residuals
Similarity Gain	Sum of the scores of the two sibling leaves minus the score of the parent node
O^t	Optimal node value at time step t
Residuals	Difference between the observed and predicted values in a model

Figure 2: XGBoost Variables and Residuals

All boosting models are based on Taylor series approximations; XGBoost, for example, employs a second-order Taylor series approximation to compute the optimal node output as $O^{(t)} \approx \frac{\text{Sum of Residuals}}{\text{Number of Residuals} + \lambda}$. It evaluates

the benefit of each split via a Similarity Score, $\mathcal{S} = \frac{(\text{Sum of Residuals})^2}{\text{Number of Residuals} + \gamma}$, and prunes branches based on the Similarity Gain, defined as $\mathcal{G} = \mathcal{S}_{\text{Child 1}} + \mathcal{S}_{\text{Child 2}} - \mathcal{S}_{\text{Parent Node}}$, with pruning occurring when $\mathcal{G} - \gamma \leq 0$. We use the categorical cross-entropy loss function, $\mathcal{L} = -\sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log P(y_i = c | X_i)$, where $y_{i,c}$ indicates sample i 's membership in class c and $P(y_i = c | X_i)$ is the predicted probability. The model is trained with 300 trees using a softmax output function that converts raw logits z_c into normalized probabilities according to $P(y = c | X) = \frac{\exp(z_c)}{\sum_{j=1}^C \exp(z_j)}$.

4.2.2 Optimized XGBoost Methodology

We developed a second XGBoost model using the newly processed dataset after dropping the Weight and Gender columns. In our framework, we employ Bayesian optimization via Optuna to fine-tune XGBoost hyperparameters (15). The hyperparameter tuning problem is treated as a black-box optimization task where the goal is to maximize the classification accuracy. Since evaluating the model (via cross-validation) is computationally expensive, we avoid a brute-force grid search and instead rely on Bayesian optimization to efficiently explore the hyperparameter space. We define an objective function that takes a set of hyperparameters (such as the number of trees, maximum tree depth, learning rate, subsample ratio, column sample ratio, gamma, and regularization parameters) as input. For each set of hyperparameters suggested by Optuna, we instantiate an XGBoost classifier with those parameters and evaluate its performance using 5-fold cross-validation on the training data. The mean accuracy obtained from cross-validation serves as the objective function's output. This effectively measures how well a particular configuration performs without needing to retrain on the full dataset. We define the following hyperparameter ranges:

Hyperparameter	Type	Range/Distribution
n_estimators	Integer	100 to 500
max_depth	Integer	3 to 10
learning_rate	Float (log scale)	10^{-3} to 10^{-1}
subsample	Float	0.5 to 1.0
colsample_bytree	Float	0.5 to 1.0
gamma	Float	0 to 5.0
reg_alpha	Float	0 to 1.0
reg_lambda	Float	1.0 to 5.0

Table 3: Hyperparameter Ranges for XGBoost Bayesian Optimization

We chose the Tree-structured Parzen Estimator to be our surrogate model, a nonparametric, tree-based method for Bayesian optimization (11). TPE efficiently handles high-dimensional search spaces and can work with both continuous and discrete parameters, making it well-suited for complex models. Furthermore, TPE requires lower function evaluations compared to grid or random search because it models the probability density of promising hyperparameters and focuses the search accordingly. TPE models the conditional probability densities $p(x|y)$, where y represents the observed performance (accuracy). It separates the observations into two groups: $l(x) = p(x|y < y^*)$: the density of hyperparameters that yield good performance
 $g(x) = p(x|y \leq y^*)$: the density of hyperparameters that yield poorer performance
 where x represents the set of hyperparameters being optimized, y being our performance metric (validation accuracy), y^* representing the threshold performance value that separates good and poor hyperparameter configurations. The $l(x)$ and $g(x)$ densities are updated through kernel density estimation, which allows TPE to increasingly focus on regions of the hyperparameter space that yield better performance.

We ran a study with 100 trials, each time proposing a new set of hyperparameters based on the surrogate model and acquisition function. The accuracy is evaluated for each set, and the surrogate model is updated with the new observation. The hyperparameters yielding the highest cross-validation accuracy are shown below:

Hyperparameter	Optimal Value	Interpretation
n_estimators	259	Moderate ensemble size that builds predictions incrementally.
max_depth	10	Allows for complex interactions but may risk overfitting if not regularized.
learning_rate	0.07	Moderate-to-low step size ensuring stable, gradual convergence while reducing overfitting risk.
subsample	0.8	Uses 80% of samples per tree, introducing randomness to help prevent overfitting.
colsample_bytree	0.9	90% of features are considered per tree, reducing variance without losing too much information.
gamma	0.0	No minimum loss reduction required for splits, indicating no additional constraint on tree splitting.
reg_alpha	0.1	Mild L1 regularization that encourages sparsity in the model weights.
reg_lambda	1.5	Moderate L2 regularization that penalizes large weights to control model complexity.

Table 4: XGBoost Bayesian Optimized Hyperparameters

These parameters collectively balance model flexibility and generalization, mitigating overfitting while capturing intricate feature relationships.

We trained the model with a categorical cross-entropy loss function and softmax output function.

4.2.3 XGBoost Performance Metrics

We first evaluated our Optimized XGBoost model by generating the following classification report:

Class	Precision	Recall	F1-Score	Support
Insufficient Weight	0.87	0.87	0.87	54
Normal Weight	0.68	0.72	0.70	58
Obesity Type I	0.78	0.89	0.83	70
Obesity Type II	0.97	0.95	0.96	60
Obesity Type III	1.00	0.98	0.99	65
Overweight Level I	0.89	0.71	0.79	58
Overweight Level II	0.86	0.86	0.86	58
Accuracy	–	–	0.86	423
Macro Avg	0.86	0.85	0.86	423
Weighted Avg	0.86	0.86	0.86	423

Table 5: Optimized XGBoost Classification Report

Precision (P) measures the proportion of correctly predicted positive cases: $P = \frac{TP}{TP+FP}$, where TP and FP are true and false positives. High precision minimizes false positives, crucial in high-cost misclassifications.

Recall (R) is the proportion of actual positive cases correctly predicted: $R = \frac{TP}{TP+FN}$, where FN is false negatives. High recall ensures most positive cases are identified, critical when missing positives is problematic.

F1-Score balances precision and recall: $F1 = 2 \times \frac{P \times R}{P+R}$. It is useful when false positives and false negatives are equally important.

Macro Average computes the unweighted mean across all classes: $\text{Macro Avg} = \frac{1}{N} \sum_{i=1}^N M_i$, treating all classes equally.

Weighted Average adjusts for class imbalance: $\text{Weighted Avg} = \frac{\sum_{i=1}^N (M_i \times S_i)}{\sum_{i=1}^N S_i}$, where S_i represents class support.

The classification report evaluates model performance across BMI-based weight categories. The model excels in

Obesity Type II and III, achieving high precision (0.97, 1.00) and recall (0.95, 0.98), meaning predictions are highly accurate. However, Normal Weight classification shows lower precision (0.68) and recall (0.72), indicating challenges in distinguishing normal-weight individuals.

Overall, the model attains 86% accuracy on 423 test samples. Macro averages slightly underperform weighted averages, reflecting class imbalance. While obesity detection is strong, improvements are needed for Normal Weight and Overweight Level I and II classifications.

We further analyzed our XGBoost model with the Confusion Matrix displayed below:

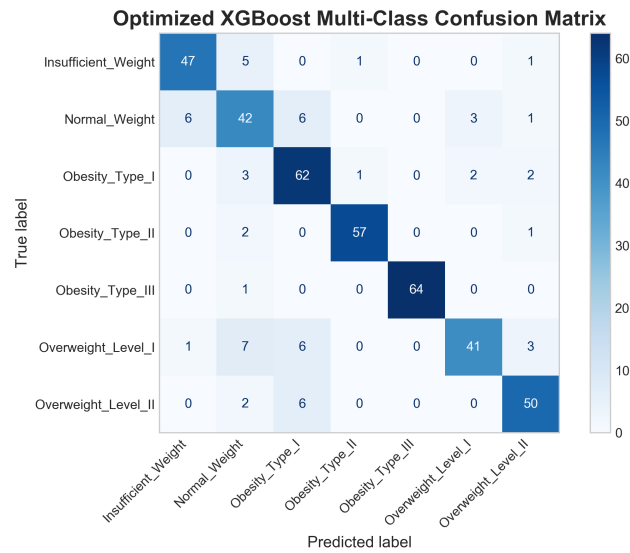


Figure 3: Optimized XGBoost Confusion Matrix

The multi-class confusion matrix shows strong performance in extreme classes (Obesity_Type_III and Obesity_Type_II), where most instances are correctly classified. However, Normal_Weight, Insufficient_Weight, and Overweight levels have lower classification accuracy.

To further evaluate model robustness, we use One-vs-Rest ROC (Receiver Operating Characteristic) curves to assess its discriminative ability. This involves binarizing each class label and applying thresholds to convert continuous probability predictions into binary predictions. At each threshold, true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN) are computed to determine the True Positive Rate (TPR) and False Positive Rate (FPR).

$$\text{True Positive Rate (TPR)} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{False Positive Rate (FPR)} = \frac{\text{FP}}{\text{FP} + \text{TN}}$$

We then calculate the per-class One-vs-Rest AUC-ROC (Area Under the ROC) and graph the ROC curve based on the outputs of these equations:

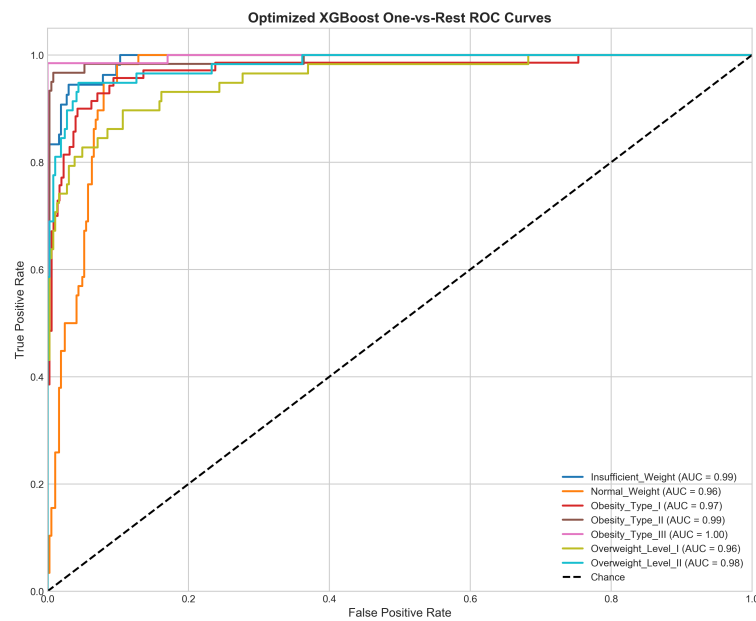


Figure 4: Optimized XGBoost AUC-ROC Curves

AUC is computed as the integral of the True Positive Rate (TPR) with respect to the False Positive Rate (FPR):

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR}) d(\text{FPR})$$

The chance curve represents if the model were to randomly guess each class when binarized, representing an AUC of 0.5. The greater the AUC and sharper the curve, the more robust it performs in differentiating that class from other classes. Most notably, the model achieved an AUC of 1.00 for Obesity Type III and an AUC of 0.99 for Obesity Type II and Insufficient Weight classes.

We also extract the feature importance of the tabular features for the XGBoost model as displayed below:

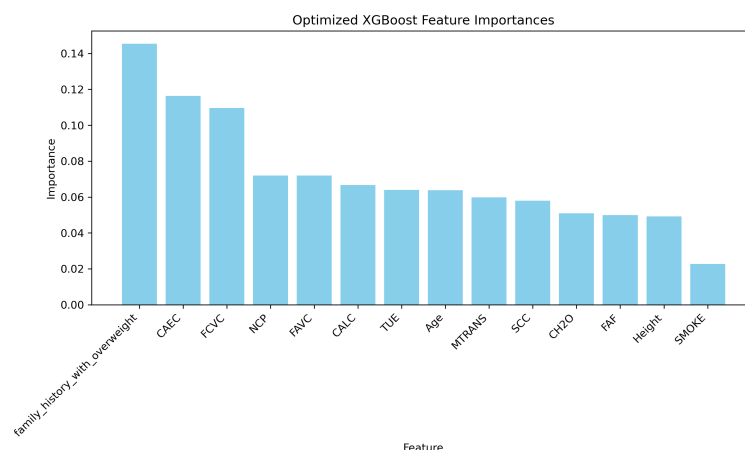


Figure 5: Optimized XGBoost Feature Importances

Interestingly, the family history feature that was originally the least informative parameter is now the most important. This is due to the elimination of overly powerful predictive features that reduces the predictive relevance of other features that hold more importance to our financial model. However, family history of obesity often indicates a genetic tendency for increased weight gain, which is not a controllable factor that we can make

recommendations for. However, the use of this feature greatly improves model robustness and predictive power.

A moderately close second to family history is the frequency people eat snacks in between meals (CAEC), followed closely by the frequency of vegetable consumption (FCVC). The next most important feature is the number of main meals (NCP) with a moderately larger dropoff in feature importance.

4.2.4 FTTransformer Architecture and Methodology

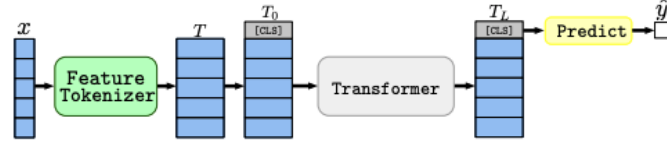


Figure 6: FTTransformer Overview

A Feature-Tokenzer Transformer is a neural network designed for tabular data, combining a feature-tokenizer with transformer layers (13). The feature-tokenizer converts variables into dense embeddings, similar to text processing in language models. These embeddings are then processed by self-attention mechanisms in transformer layers, dynamically modeling feature interdependencies.

After transformation, tokenized categorical features are fused with tokenized continuous ones to create a comprehensive representation, which is fed into multilayer perceptrons for classification.

FTTransformer excels in capturing complex feature interactions but is computationally intensive compared to traditional models like XGBoost. Its self-attention mechanism adds interpretability challenges, and its complexity can lead to overfitting, especially in smaller datasets or when inter-feature relationships are less intricate.

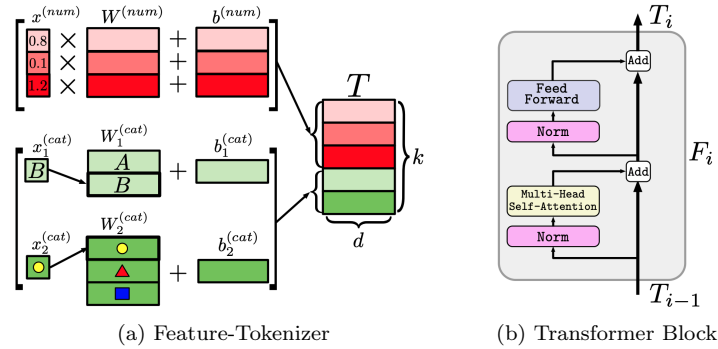


Figure 7: Parts of Feature-Tokenzer Transformer

4.3 Feature Tokenization and Transformer Blocks

Feature Tokenization

Numerical and categorical features are combined into a unified tensor T via tokenization. For numerical features $\{x_1^{(num)}, \dots, x_N^{(num)}\}$, each feature is linearly embedded as

$$T_j^{(num)} = b_j^{(num)} + x_j^{(num)} \cdot W_j^{(num)} \in \mathbb{R}^d,$$

where $W_j^{(num)}$ and $b_j^{(num)}$ are learned parameters. For categorical features $\{x_1^{(cat)}, \dots, x_M^{(cat)}\}$, each is embed-

ded via

$$T_j^{(cat)} = b_j^{(cat)} + e_j^T W_j^{(cat)} \in \mathbb{R}^d,$$

with e_j^T a one-hot vector. The tokens are fused by stacking:

$$T_0 = \text{stack} \left[T_1^{(num)}, \dots, T_N^{(num)}, T_1^{(cat)}, \dots, T_M^{(cat)} \right] \in \mathbb{R}^{(N+M) \times d}.$$

Transformer Blocks

Each Transformer block refines token embeddings T_{i-1} via:

1. **LayerNorm:** Applied to T_{i-1} as

$$T_{i-1} = \text{LayerNorm}(T_{i-1}),$$

normalizing each token x via

$$\hat{x} = \frac{x - \mu}{\sigma} \gamma + \beta.$$

2. **Multi-Head Self-Attention:** T_{i-1} is projected into queries Q , keys K , and values V . For each head,

$$\text{head}_i = \text{softmax}\left(\frac{Q_i K_i^T}{\sqrt{d_h}}\right) V_i,$$

and concatenated with a final projection:

$$\text{MultiHead}(T_{i-1}) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O.$$

3. **Residual & LayerNorm:** The attention output is added to T_{i-1} and normalized:

$$T'_i = \text{LayerNorm}(T_{i-1} + \text{MultiHead}(T_{i-1})).$$

4. **Feed-Forward Network (FFN):** Each token is processed as

$$\text{FFN}(x) = \sigma(xW_1 + b_1)W_2 + b_2.$$

5. **Residual:** Finally,

$$T_i = T'_i + \text{FFN}(T'_i).$$

We fine-tuned the FTTransformer using a Bayesian optimization procedure over 50 trials with the same TPE framework and additionally incorporated trial pruning to terminate unpromising trials early in the optimization process in order to save computational resources and time as a result of the more complex architecture. After each epoch in the trial, our optimizer compares validation loss against the best performance observed so far. If the trial's performance is significantly worse than expected, the trial is pruned (halted early), which allows Bayesian optimization to explore a larger hyperparameter space much faster.

Hyperparameter	Type	Range/Distribution
dim	Categorical	32, 64, 128
depth	Integer	2 to 5
heads	Integer	2 to 8
attn_dropout	Float	0.1 to 0.5
ff_dropout	Float	0.1 to 0.5
learning_rate	Float (log scale)	10^{-4} to 10^{-3}
weight_decay	Float (log scale)	10^{-7} to 10^{-5}
num_epochs	Integer	10 to 50

Table 6: Hyperparameter Ranges for TabTransformer Bayesian Optimization

Our Bayesian optimization study reached the following hyperparameters:

Hyperparameter	Optimal Value	Interpretation
dim	128	Embedding dimension for categorical features. A lower dimension can reduce overfitting and computation.
depth	3	Number of transformer layers. A shallow architecture reduces complexity and speeds up training.
heads	8	Number of attention heads. Sufficient for multi-view attention while keeping computations low.
attn_dropout	0.24202	Moderate dropout in attention layers to prevent overfitting.
ff_dropout	0.39235	Slightly higher dropout in feed-forward layers for additional regularization.
learning_rate	0.00048	A low learning rate ensures gradual convergence and model stability.
weight_decay	5.13e-07	Very small L2 regularization to lightly penalize large weights.
num_epochs	26	Number of training epochs to balance learning and generalization.

Table 7: Optimized FTTransformer Hyperparameters

We then trained a separate FTTransformer using these hyperparameters, the Adaptive Moment Estimator optimizer, the categorical cross-entropy loss function, and a softmax output function (16).

4.3.1 FTTransformer Results

Our model achieved an 82% testing accuracy with the following supporting metrics obtained from running our FTTransformer on the testing set:

Class	Precision	Recall	F1-Score	Support
Insufficient Weight	0.82	0.91	0.86	54
Normal Weight	0.64	0.59	0.61	58
Obesity Type I	0.86	0.81	0.84	70
Obesity Type II	0.95	0.90	0.92	60
Obesity Type III	0.97	0.98	0.98	65
Overweight Level I	0.79	0.76	0.77	58
Overweight Level II	0.69	0.78	0.73	58
Accuracy	–	–	0.82	423
Macro Avg	0.82	0.82	0.82	423
Weighted Avg	0.82	0.82	0.82	423

Table 8: Optimized Classification Report for Multi-Class BMI Prediction

The FTTransformer model achieved strong overall performance in multi-class obesity classification, with a macro F1-score of 81.63% and a weighted F1-score of 81.97%, indicating balanced results across classes and robust handling of frequent categories. Notably, the model excels in detecting severe obesity, with F1-scores of 92.31% (Obesity Type II) and 97.71% (Obesity Type III), and nearly perfect recall for Obesity Type III (98.46%). It also performs well for Obesity Type I (83.82%) and Insufficient Weight (85.96% F1, 90.74% recall). However, classification of Normal Weight (F1 = 61.26%, precision = 64.15%, recall = 58.62%) and Overweight Level II (F1 = 73.17%) remains challenging, likely due to misclassification among similar body composition categories. A confusion matrix generated on the testing set further details these classification errors. Future work should address these issues through enhanced feature engineering, data augmentation, and the inclusion of additional biomarkers.

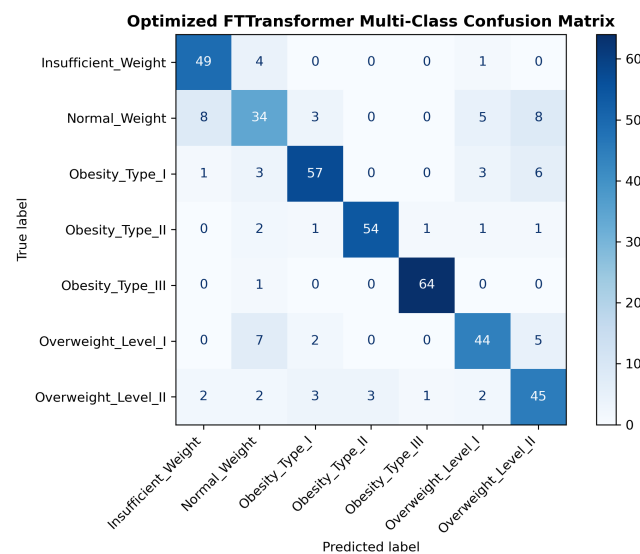


Figure 8: Optimized FTTransformer Confusion Matrix

The confusion matrix shows that FTTransformer excels in severe obesity detection (64/65 for Obesity Type III, 54/60 for Type II, and 57/70 for Type I) and performs well for Insufficient Weight (49/54 correct), but it struggles with Normal Weight and Overweight Levels I and II (only 34/58 Normal Weight cases correct), with many misclassifications between the overweight categories due to feature overlap.

To evaluate the discriminative ability of the FTTransformer model across the seven weight categories, we generated one-vs-rest Receiver Operating Characteristic curves (17).

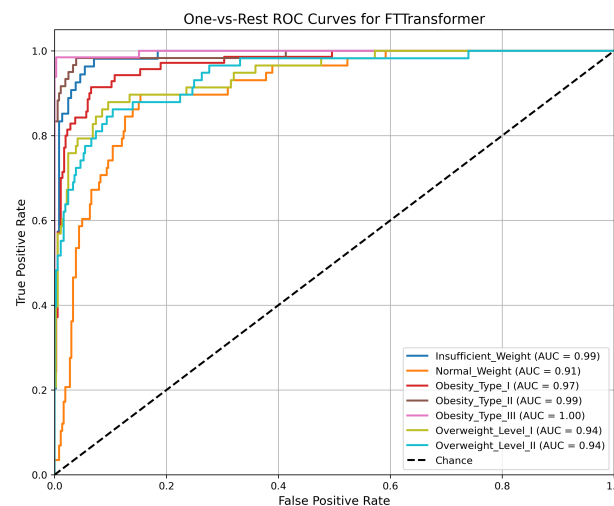


Figure 9: Optimized FTTransformer ROC Curve

The Receiver Operating Characteristic (ROC) curves illustrate the FTTransformer model's one-vs-rest classification performance across seven weight categories. The ROC curve evaluates the trade-off between the true positive rate (sensitivity) and the false positive rate, with curves closer to the upper-left corner indicating better classification performance.

The Area Under the Curve (AUC) values measure the model's ability to distinguish between classes, with 1.0 indicating perfect classification and 0.5 representing random guessing. All classes achieve high AUC scores (above 0.96), suggesting strong separability. Obesity Type III reaches an AUC of 1.00, while Insufficient Weight and Obesity Type II achieve 0.99, indicating near-perfect classification. Normal Weight and Overweight Level I

have slightly lower AUC values (0.96), reflecting some overlap in BMI-related features, while Overweight Level II achieves 0.98.

All ROC curves remain well above the random classification baseline ($AUC = 0.5$), confirming the model's strong discrimination ability. The sharp rise in most curves indicates high sensitivity with minimal false positives, which is critical in medical classification tasks. Additionally, training loss and accuracy curves track the model's learning behavior and generalization performance over multiple epochs.

To assess the learning behavior and generalization ability of the FTTransformer model, we monitored its training progress using loss and accuracy curves over multiple epochs. These graphs provide insights into how well the model adapts to the dataset during training and how effectively it performs on unseen validation data.

4.4 Logistic Classification

We used Multinomial Logistic Classification as a baseline model to establish a benchmark for evaluating more complex models like XGBoost and FTTransformer (18). Logistic Classification serves as a baseline because it is a simple, interpretable, and widely used classification model that assumes a linear decision boundary between classes. Since it lacks the complexity of tree-based and transformer-based models, its performance provides a reference point for assessing improvements.

We did not include more in-depth metrics like the confusion matrix or ROC analysis because Logistic Classification's relatively low performance (53% accuracy) suggests it is not a strong candidate for deployment. Instead, we focused on high-level metrics (precision, recall, and F1-score) to compare overall model effectiveness without excessive analysis of a model we are not advancing to later phases.

4.4.1 Logistic Classification Architecture

Logistic Classification is a linear classification model that predicts the probability of an instance belonging to a specific class. In multinomial Logistic Classification, the model extends beyond binary classification by using the softmax function to assign probabilities across multiple classes.

The implementation uses the L-BFGS solver (Limited-memory Broyden-Fletcher-Goldfarb-Shanno), an efficient quasi-Newton method that approximates the Hessian matrix to optimize the loss function (19). L-BFGS is well-suited for large datasets due to its ability to handle high-dimensional parameter spaces with limited memory requirements.

We used the categorical cross-entropy loss function with 500 iterations to train this model.

The model achieved an overall accuracy of 53%, indicating limited predictive power compared to more advanced models like XGBoost and FTTransformer.

4.5 Model Comparison

To evaluate multi-class obesity classification, we compared XGBoost, FTTransformer, and Multinomial Logistic Classification using accuracy, precision, and recall, with a focus on macro and weighted averages. The baseline Logistic Classification model provides a reference, while FTTransformer captures complex feature interactions, and XGBoost leverages gradient boosting for structured data. The analysis assesses each model's strengths and selects the best-performing one for Phase 3.

Model	Accuracy	Macro Avg Precision	Macro Avg Recall	Weighted Avg Precision	Weighted Avg Recall
XGBoost	85.82%	86.00%	85.00%	86.00%	86.00%
FTTransformer	82.00%	81.67%	81.81%	82.12%	82.03%
Logistic Classification	53.00%	52.00%	52.00%	52.00%	53.00%

Table 9: Comparison of Model Performance on Multi-Class Obesity Classification

The baseline Multinomial Logistic model achieved 53.00% accuracy with poor recall for intermediate weight classes. The FTTransformer improved accuracy to 82.00% (macro precision/recall 81.7%), but struggled with

Normal and Overweight distinctions. In contrast, XGBoost achieved 85.82% accuracy and around 86% precision/recall, excelling in severe obesity detection and reducing misclassifications among adjacent BMI groups. Given its balanced performance and superior accuracy, XGBoost is selected as the final model for further optimization.

4.6 Phase 3: Per-Sample Monte Carlo Risk Mitigation Optimization

Monte Carlo simulation is a computational technique that uses repeated random sampling to obtain numerical results (20). It is often used to model phenomena with significant uncertainty in their inputs, where analytical solutions are difficult or impossible to derive. By simulating a large number of scenarios, Monte Carlo simulation provides insight into the possible variability in model predictions, offering a robust tool for understanding and managing uncertainty. In our approach, we generate personalized recommendations by modifying key modifiable features using a Monte Carlo simulation framework. The four modifiable features considered are:

1. FAF: Physical Activity Frequency
2. SCC: Calorie Consumption Monitoring (treated as a categorical variable)
3. CAEC: Consumption of Food Between Meals
4. CH2O: Consumption of Water in Liters

We chose these specific aspects due to their feature importance in our XGBoost analysis, their controllability (for example, family history is significant but not within the individual's control), and their overall feasibility for real-world implementation.

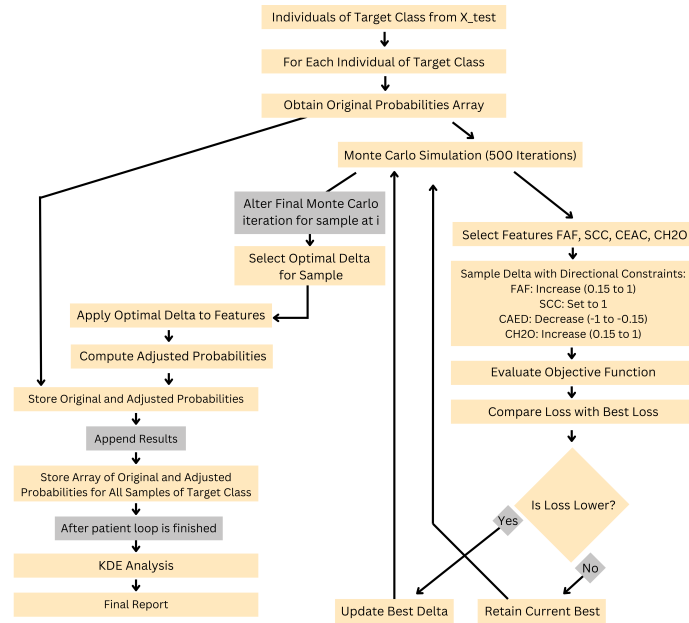


Figure 10: Monte Carlo Methodology Overview

4.6.1 Part 1: Obesity

We use Monte Carlo simulation to generate personalized recommendations for obese individuals (Obesity Types I–III) using an optimized XGBoost model. The model outputs class probabilities, with overall obesity probability defined as the sum for the three obesity classes. Four lifestyle features—Physical Activity Frequency (FAF), Calorie Consumption Monitoring (SCC), Consumption of Food Between Meals (CAEC), and Daily Water Consumption (CH2O)—are adjusted to reduce obesity risk while limiting underweight risk. Interventions are

evaluated using the loss function

$$\text{loss}(\delta) = \log(p_{\text{obesity}}(\delta) + \epsilon) + \alpha \max(0, p_{\text{insufficient}}(\delta) - T)^2 + \lambda \|\delta\|^2,$$

with $\epsilon = 10^{-8}$, $T = 0.1$, $\alpha = 100$, and $\lambda = 0.1$. For each individual, 500 Monte Carlo samples are drawn from $\mathcal{N}(0, 0.2)$ and constrained such that FAF increases (0.15–1), SCC is set to 1, CAEC decreases (–1 to –0.15), and CH2O increases (0.15–1). The optimal modification minimizes the loss and is applied to compute post-intervention obesity probabilities. Our intervention reduced obesity probability by 36.14% (CI: 33.14–39.14%), confirming the importance of exercise, calorie tracking, reduced snacking, and hydration in effective weight management. For further insights, see Section 6: Recommendations.

KDE is a non-parametric technique used to estimate the probability density function (PDF) of a dataset (21). Unlike histogram-based methods, KDE produces a continuous and smoothed density function, which makes it useful for analyzing the distribution of data without relying on predefined bin widths. The estimated density at point x is given by:

$$f(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right)$$

where n is the number of data points, $K(\cdot)$ is the kernel function, and h is the bandwidth parameter, which controls the smoothness of the density estimate. We generated a kernel density estimate (KDE) plot to visualize the distribution of observations in the dataset before and after the applied recommendations.

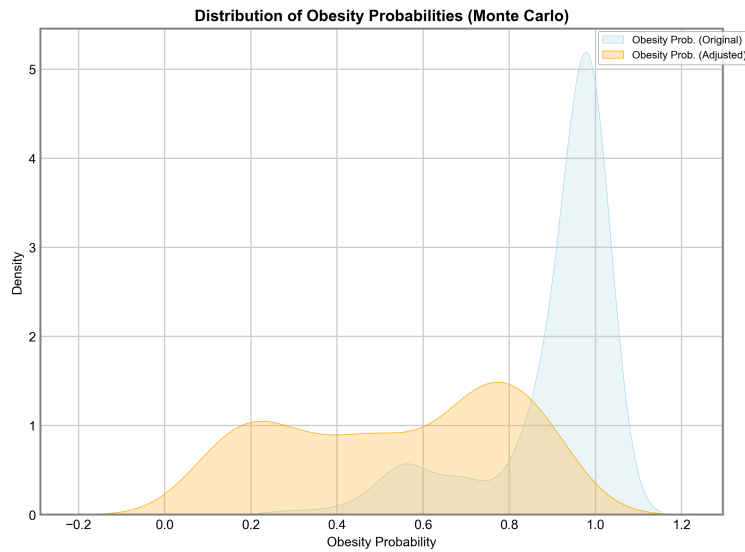


Figure 11: Obesity KDE Graph

Before intervention, the probability distribution is sharply concentrated near 1.0, indicating high confidence in obesity classification across most individuals. This suggests minimal variation in obesity risk within the population.

After intervention, the distribution shifts significantly toward lower values, with increased density in lower probability ranges. This reflects a substantial reduction in obesity risk due to personalized modifications in physical activity (FAF), calorie monitoring (SCC), snacking habits (CAEC), and water intake (CH2O).

While some individuals still have high obesity probabilities, the overall distribution is more balanced, with many shifting from high to moderate or low-risk classifications. This highlights the effectiveness of the intervention in reducing obesity probability across the population.

4.6.2 Part 2: Overweight

Our Monte Carlo simulation for risk mitigation strategies for overweight individuals follows a similar architecture as the one described above. However, it includes two key differences: it adds the two overweight classes (Overweight Level I and Level II), and it adds an extra term to the objective function that penalizes the loss for increasing the probability of obesity.

We defined the new equation as:

$$\text{loss}(\delta) = \log(p_{\text{overweight}}(\delta) + \epsilon) + \alpha \max(0, p_{\text{obesity}}(\delta) - p_{\text{obesity, orig}})^2 + \beta \max(0, p_{\text{insufficient}}(\delta) - p_{\text{insufficient, orig}})^2 + \lambda \|\delta\|^2$$

The recommended modifications for overweight individuals followed a consistent trend, reducing overweight probability by 13.5% (CI: 9.8% to 17.19%) and reducing obesity probability from 10.5% to 7.9% (24.8% decrease). Physical activity (FAF) increased (0.15–0.61), reinforcing exercise’s role in weight management. Calorie monitoring (SCC) was universally set to 1.00, emphasizing its importance in maintaining a healthy weight. Snacking (CAEC) decreased (-0.22 to -0.15), encouraging reduced calorie intake, while water consumption (CH2O) increased (0.15–0.64), supporting metabolic regulation. For further insights, see Section 6: Recommendations.

We generated a Kernel Density Estimation (KDE) graph to assess the reduction in overweight probability achieved through the recommended interventions.

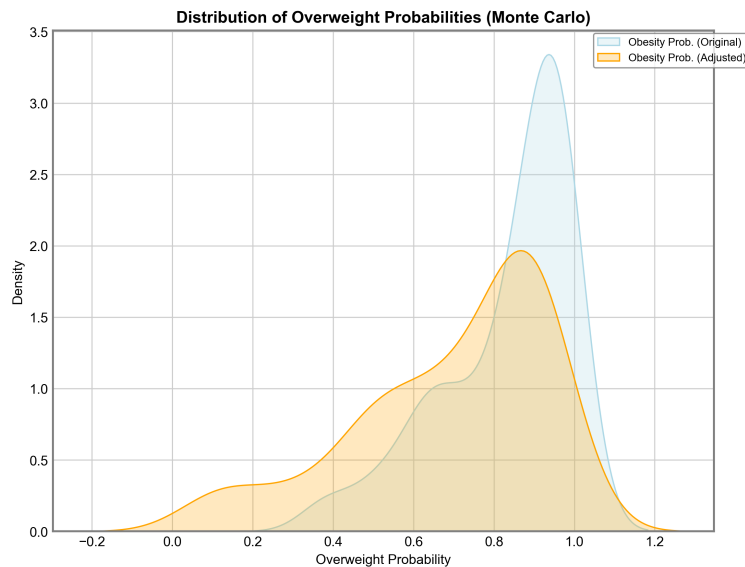


Figure 12: Obesity KDE Graph

Before the intervention, the original probability distribution is concentrated near 1.0, suggesting that most overweight individuals had a high likelihood of being classified as overweight with minimal variation. This indicates that the predictive model was highly confident in their overweight classification.

Following the intervention, a clear shift in the probability distribution is observed, with the peak moving to lower probability values. The adjusted distribution is broader and exhibits a higher density at moderate probability levels (0.4–0.8), indicating that a substantial portion of individuals experienced a reduction in their overweight classification probability. This shift suggests that the personalized lifestyle adjustments—including increased physical activity (FAF), mandatory calorie monitoring (SCC), reduced snacking (CAEC), and improved hydration (CH2O)—were effective in lowering overweight risk across the population.

Notably, while some individuals still have higher overweight probabilities, the overall trend demonstrates a significant redistribution toward lower values, reinforcing the effectiveness of targeted behavioral interventions. The increased variability in the adjusted distribution suggests that the impact of the interventions varied among individuals, with some achieving greater reductions in overweight probability than others.

4.7 Markov Chain: Risk Mitigation Impact Modeling

To analyze long-term weight category transitions, we implemented a discrete-time Markov Chain with four states (22): Insufficient Weight (IW), Normal Weight (NW), Overweight (OW), and Obese (OB). The population distribution across these states evolves according to the transition probability matrix P , where each entry P_{ij} represents the probability of moving from state i to state j in a given time step. The weight distribution at time $t + 1$ is computed as:

$$\mathbf{w}(t + 1) = \mathbf{w}(t)P,$$

where $\mathbf{w}(t)$ is the state distribution vector at time t .

4.7.1 Baseline Transition Matrix Estimation

The baseline transition matrix P was estimated using weight classification data from the dataset (1988–2018) (3). The Insufficient Weight variable was missing, so generated the distribution history for this class by assuming a negative linear relationship starting from a proportion of 0.023 in 1988 to a proportion of 0.016 in 2018. We aggregated historical cross-sectional distributions into four weight categories (Insufficient Weight, Normal Weight, Overweight, Obese) and estimated transition probabilities by fitting a Markov model to match observed weight shifts over time. The optimization objective function was formulated as follows.

$$\min_T \sum_t \left\| \mathbf{w}(t + 1) - \mathbf{w}(t)T^{\Delta t} \right\|^2 + \lambda \sum_i \left(\sum_j T_{ij} - 1 \right)^2 + \gamma \max(0, \tau - T_{OW,OB})^2,$$

where:

- $T^{\Delta t}$ is the transition matrix raised to the power of the time interval Δt ,
- The first term minimizes the difference between observed and predicted state distributions,
- The second term enforces probability row sums to be 1,
- The third term penalizes transition probabilities from overweight to obesity that fall below a threshold τ .

The optimized yearly Markov Transition Matrix P was obtained using constrained minimization, where all elements were bounded between $[0,1]$ and rows were constrained to sum to 1. The resulting Transition Matrix is:

$$P = \begin{bmatrix} 0.9803 & 0.0184 & 0.0012 & 0.0001 \\ 0.0062 & 0.9407 & 0.0498 & 0.0033 \\ 0.0009 & 0.0423 & 0.8876 & 0.0692 \\ 0.0001 & 0.0021 & 0.0514 & 0.9464 \end{bmatrix}$$

It represents the probability for every class to change into another class. For example, every year, an Overweight individual has an 88.76% change of staying Overweight, but a 6.92% chance of becoming Obese.

4.7.2 Intervention-Adjusted Transition Model

To evaluate the impact of behavioral and lifestyle interventions, we introduced an adjusted transition matrix P' , modified by empirically derived intervention effects. These adjustments were applied to transitions leading to overweight and obesity:

$$P' = P \circ M,$$

where M is a matrix of intervention-based adjustment multipliers, and $\circ$ denotes the Hadamard (element-wise) product.

The following modifications were applied to simulate the effects of interventions:

- Decreased Normal Weight to Overweight conversion by 15%

- Increased Overweight to Normal Weight conversion by 13.5%
- Decreased Overweight to Obese conversion by 24.8%
- Increased Obese to Overweight conversion by 36.14%

Starting from the baseline transition matrix P , which was calibrated using cross-sectional survey data, we introduced intervention effects by defining a modifiers matrix. This modifiers matrix applied element-wise adjustments to specific transition probabilities, as displayed below:

$$P' = \begin{bmatrix} 0.4118 & 0.2689 & 0.3193 & 0.0000 \\ 0.0429 & 0.1195 & 0.8333 & 0.0043 \\ 0.0007 & 0.7629 & 0.1926 & 0.0438 \\ 0.0003 & 0.0000 & 0.0583 & 0.9414 \end{bmatrix}$$

while increasing the reverse transitions—to account for behavioral and outcome modifications. After applying these modifiers, the resulting matrix P_{mean} was normalized so that each row summed to 1. Next, we defined an initial population distribution

$$\mathbf{w}(0) = \begin{bmatrix} 0.016 \\ 0.295 \\ 0.307 \\ 0.403 \end{bmatrix},$$

which corresponds to 1.6% Insufficient Weight, 29.5% Normal Weight, 30.7% Overweight, and 40.3% Obese, and normalized it to sum to 1. The evolution of the weight distribution over time was then simulated using the discrete-time Markov chain model. Specifically, for each subsequent year t , the distribution was updated via

$$\mathbf{w}(t) = \mathbf{w}(t-1) P_{\text{mean}}.$$

with a 20-year time span. For final results, see Recommendations 6.4.3.

5 Risk Analysis

5.1 Defining the Risks

Obesity poses multifaceted risks that span across both health and economic dimensions.

5.1.1 Personal Health Risks

Obesity increases the risk of life-altering conditions such as cardiovascular diseases (stemming from high blood pressure, high cholesterol, and type 2 diabetes), various cancers, mental health disorders, respiratory issues like asthma and sleep apnea, and joint problems (8). Obesity is associated with one out of every three heart attack or stroke deaths and one in twelve cancer deaths (23). Moreover, life expectancy is significantly affected—where individuals with Class I and Class II obesity could lose between 2 and 4 years of life and individuals with Class III obesity could lose up to 14 years (4; 24).

5.1.2 Societal and Economic Risks

Each adult with obesity faces an excess annual healthcare costs of about \$1,861, and each overweight adult incurs an annual excess medical cost of around \$600 (25). In total, the CDC estimates the annual national cost of obesity to be around \$173 billion a year (1).

Beyond these direct costs, obesity also has large, indirect economic impacts. On average, obesity reduces life expectancy by roughly 4.7 years per person, resulting in a total loss of about 515 million years of life—which is comparable to the complete lifetimes of approximately 6.75 million people (26). This reduction in life expectancy effectively reduces the labor supply; specifically, it is expected to cost the U.S. economy approximately \$5.6 trillion in potential GDP (short for Gross Domestic Product, measured annually in America) and result in about \$1.0 trillion in lost Federal tax receipts over the next decade (26). Moreover, obesity leads to a decline in labor

productivity through increased absenteeism and diminished on-the-job output, resulting in upwards of \$6.38 billion GDP (8).

5.1.3 Sensitivity Analysis

The sensitivity analysis aims to evaluate the impact of incremental modifications to key modifiable features—FAF (physical activity frequency), CH2O (daily water consumption), and CAEC (consumption of food between meals)—on the predicted probabilities of different weight categories. This analysis examines how adjusting these features, individually and in combination, influences classification probabilities for four target groups: Insufficient Weight, Normal Weight, Overweight, and Obese. A structured grid of values ranging from -0.8 to 0.8 is added to each modifiable feature, systematically increasing or decreasing the feature's value while ensuring that it remains within its observed range. For each modified input, the trained classification model predicts probability distributions across the seven weight classes, and the mean probability for the relevant group is computed across all individuals. Additionally, a composite sensitivity analysis is performed, where all features are adjusted simultaneously, but CAEC is modified in the opposite direction to reflect its inverse relationship with weight status. This systematic approach allows for a direct comparison of how lifestyle modifications influence classification outcomes, providing insights into which features have the most significant impact on weight predictions.

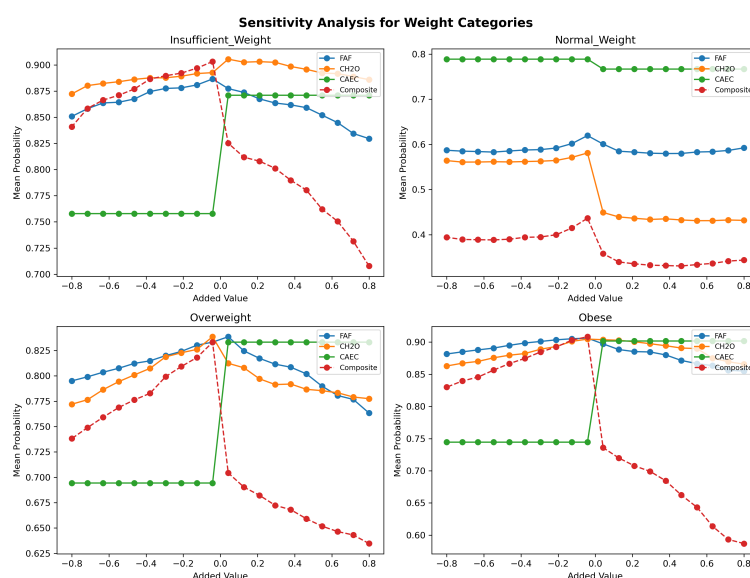


Figure 13: Sensitivity Analysis

The sensitivity analysis revealed distinct response patterns across weight categories when modifiable features—FAF (physical activity), CH2O (water intake), and CAEC (snacking frequency)—were adjusted.

For **Insufficient Weight**, small FAF changes had little effect, but a significant decrease sharply lowered underweight probability. CH2O and CAEC remained stable except at larger increments. Simultaneous feature modifications followed a similar trend, with a sharp probability drop after a moderate negative shift.

In the **Normal Weight** category, changes in FAF, CH2O, and CAEC produced gradual probability shifts, with no abrupt transitions. The composite curve showed a smooth trajectory, indicating greater adaptability to moderate lifestyle changes.

For **Overweight**, classification probability was high at baseline, suggesting a strong model bias. FAF and CH2O had mild effects, while CAEC had a greater impact at certain increments, highlighting its role in overweight classification. Composite adjustments, particularly CAEC reductions, led to more pronounced probability shifts.

In the **Obese** category, FAF and CH2O had modest effects, but reducing CAEC beyond a threshold significantly lowered obesity probability. The composite curve reinforced this trend, showing a sharp probability decline for moderate negative increments.

Overall, **feature sensitivity varied by weight category**. Insufficient Weight and Obese groups exhibited abrupt probability transitions, while Normal Weight showed gradual responses. Overweight classification remained stable for individual changes but responded more strongly to composite modifications. CAEC played the dominant role in overweight and obese classifications, with its reduction leading to significant probability drops. FAF and CH2O had more gradual effects, though a sharp FAF reduction could drastically alter Insufficient Weight classification. Simultaneous adjustments consistently amplified probability shifts, underscoring the importance of coordinated lifestyle changes in weight classification predictions.

5.1.4 Groups at Risk

Obesity can affect anyone, but it is more prevalent in certain demographic groups. According to the CDC, 40.3% of American adults were classified as obese and 9.7% as severely obese between 2021 and 2023 CDC. In contrast, about one in five American children are obese (4). Among U.S. adults, the 40-to-59 age group had the highest average rate of obesity—approaching 50%—with men and women in this age range exhibiting similar obesity rates (4).

5.1.5 Severity

We define severity for the monetary losses of being overweight and obese. Currently, a randomly selected individual in the US has a 30.7% chance of being overweight and 42.4% chance of being obese. On average, overweight individuals incur an excess of \$600 for medical costs, while individuals with obesity, on average incur an excess of \$1,816 in medical costs (25). For our insurance company of 100,000 policyholders with a baseline 70% coverage of weight-related medical costs, we calculate that the severity of our company's expenses covering the policyholders will be \$95,418,400 per year.

5.1.6 Likelihood

Our likelihood of a risk is represented by our Markov Transition Matrix. We calculated the baseline matrix to be:

$$P = \begin{bmatrix} 0.9803 & 0.0184 & 0.0012 & 0.0001 \\ 0.0062 & 0.9407 & 0.0498 & 0.0033 \\ 0.0009 & 0.0423 & 0.8876 & 0.0692 \\ 0.0001 & 0.0021 & 0.0514 & 0.9464 \end{bmatrix}$$

This matrix defines the probability of one class to change to another each year. For example, an obese individual has a 94.64% chance of remaining obese and a 5.14% chance of becoming overweight the following year. We expect the transition matrix's probability to remain the same every year. We also calculate the transition matrix for when the insurance company implements our recommendations, as seen in Recommendations 6.4.3.

5.1.7 Uncertainty

Our patient-level XGBoost and FTTransformer classification models both incorporate softmax outputs, which means that they output an array of 7 probabilities, one probability for each class. To obtain a definitive class from this uncertain output, we must apply an argmax function, which outputs the class with the greatest probability. For example, if our XGBoost model predicts that a certain sample has a probability distribution for the 7 classes of:

$$\text{softmax}(x_i) = [0.034 \quad 0.072 \quad 0.101 \quad 0.723 \quad 0.031 \quad 0.004 \quad 0.035]$$

This means that XGBoost predicts that the sample at i has a 3.4% chance of being Insufficient Weight, a 7.2% chance of being Normal Weight, a 10.1% chance of being Obesity Type I, a 72.3% chance of being Obesity Type II, a 3.1% chance of being Obesity Type III, a 0.4% of being Overweight Level I, and a 3.5% chance of being Overweight Level II. After applying argmax to our softmax output, we get the output 3, which corresponds to Obesity Type II, hence accounting for the prediction uncertainty.

5.1.8 Expected Value

We expect a large amount of policyholders to be obese and overweight by 2045. We predict that 43,524 policyholders will be obese by 2025 using our Markov Chain analysis. We also predict a slight decrease in the overweight population, where it peaks and declines to 30,037 individuals by 2045, in large part due to more overweight individuals becoming obese.

6 Recommendations

6.1 Behavior Change

Behavior change aims to prevent obesity by promoting healthier lifestyle choices through early intervention strategies, focusing on reducing obesity incidence rather than addressing individuals already classified as obese or overweight. We recommend improving physical activity, nutrition, and overall well-being through community-based programs and educational policies. Possible initiatives include incorporating daily physical activity in schools, ensuring nutritious cafeteria meals, creating accessible public parks and bike lanes, and building affordable local recreation centers.

6.2 Modifying Outcomes

Modifying outcomes focuses on reducing the severity of obesity-related conditions once an individual is already classified as obese or overweight. Unlike behavior change, which aims to prevent obesity in the general population, modifying outcomes involves structured interventions that mitigate health risks and reduce long-term healthcare costs. This study uses a data-driven Monte Carlo-based intervention framework to optimize lifestyle modifications for individuals with obesity or overweight classification.

6.2.1 Personalized Lifestyle Adjustments

Our Monte Carlo simulation identified four key lifestyle factors influencing obesity: physical activity, calorie monitoring, snacking frequency, and hydration. For obese individuals, interventions reduced obesity probability by 36.14% (95% CI: 33.14–39.14%). Recommended adjustments include increasing physical activity by 0.15–0.78 units (e.g., shifting from exercising 1–2 days to 2–4 or 4–5 days per week), ensuring daily calorie tracking (SCC = 1), reducing between-meal snacking by 0.15–0.33 units (moving from frequent to occasional snacking), and increasing water consumption by 0.15–0.63 units (e.g., from less than 1 L to 1–2 L or above). For overweight individuals, similar but less intense modifications—yielding a 13.5% reduction in overweight probability (95% CI: 9.8–17.19%) and a 24.8% decrease in obesity risk—are advised: increase physical activity by 0.15–0.61 units, maintain daily calorie monitoring, reduce snacking by 0.15–0.22 units, and boost water intake by 0.15–0.64 units. These recommendations form our outcomes plan, which is designed to improve metabolic health by shifting individuals toward healthier weight classifications.

6.3 Insurance

6.3.1 Overview

We propose an insurance plan that rewards proactive health management with dynamic incentives, focusing on structure, pricing adjustments, and coverage for obesity-related diseases rather than reiterating behavioral recommendations.

6.3.2 Incentives and Rewards Structure

Tiered Premium Discounts:

- **Baseline Tier:** Policyholders receive a 5% discount on their premium upon enrollment in the wellness program.
- **Active Engagement Tier:** Achieving predefined health milestones (e.g., regular exercise, daily calorie tracking) increases the discount to 10%.
- **Optimal Health Tier:** Consistent overachievement over a 12-month period qualifies for a maximum discount of 15%.

Cashback and Reward Bonuses:

- **Monthly Rewards:** Receive \$20 cashback when combined wellness targets (physical activity, calorie monitoring, snacking reduction, and increased hydration) are met.
- **Annual Bonus:** Achieve optimal health performance for the year to earn an additional \$150 reward.

6.3.3 Premium Structure**Base Premium:**

- The base annual premium for the plan is set at \$500.

Adjusted Premium Plans:

- **Baseline Tier (5% discount):** $\$500 - \$25 = \$475$ annual premium.
- **Active Engagement Tier (10% discount):** $\$500 - \$50 = \$450$ annual premium.
- **Optimal Health Tier (15% discount):** $\$500 - \$75 = \$425$ annual premium.

6.3.4 Coverage for Obesity-Related Diseases

The plan offers enhanced coverage for obesity-related conditions such as diabetes, hypertension, cardiovascular disease, and certain cancers:

- **Standard Coverage:** 70% of expenses for obesity-associated conditions.
- **Enhanced Coverage:** Policyholders demonstrating sustained health improvements may receive up to 80% coverage.
- **Performance Bonus:** Meeting specific annual wellness benchmarks can further boost coverage by 5%, reaching up to 85% in total.

6.3.5 Implementation and Monitoring

- Enrollment in a dedicated digital health platform is required for continuous monitoring and reporting.
- Regular assessments track progress against wellness targets, ensuring transparent adjustments to premiums and coverage.
- Quarterly progress reports are provided to policyholders, detailing their discount status, rewards earned, and current coverage enhancements.

6.3.6 Markov Chain Results and Long-Term Impact

After iterating this process over a 20-year horizon, the sequence of distributions is captured in the array $\text{dist}_{\text{mean}}$, which represents the projected evolution of the population's weight categories under the intervention-adjusted transition dynamics.

The intervention-adjusted simulation showed:

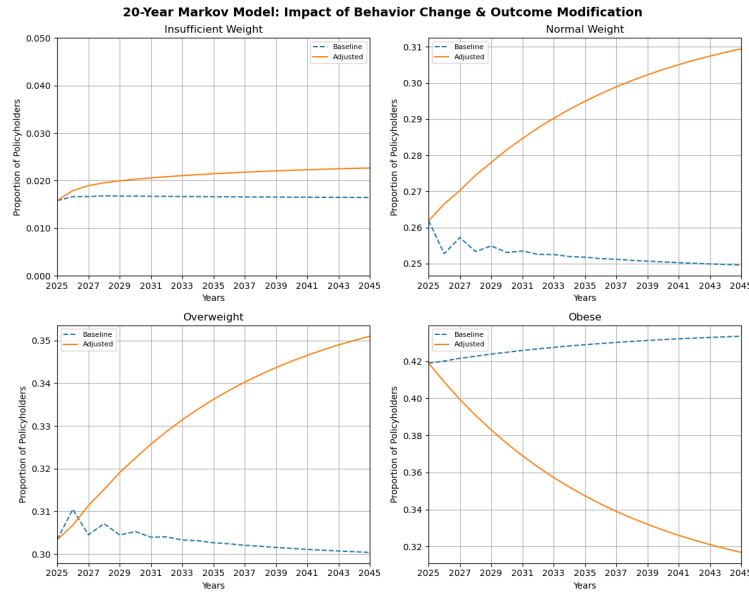


Figure 14: 20-Year Markov Chain Monte Carlo of Impacts of Recommendations

This analysis evaluates the impact of behavior change and outcome-modifying strategies on 100,000 policyholders representative of the U.S. weight distribution over 20 years. The simulation compares a baseline scenario, where obesity and overweight rates rise, with an adjusted scenario incorporating interventions to promote healthier weight transitions.

Under the baseline scenario, the insufficient weight category remains stable, increasing slightly from 1,581 to 1,647 individuals. In contrast, the adjusted model sees a more noticeable rise to 2,267 individuals, suggesting interventions prevent excessive weight gain. The normal-weight category declines in both cases, from 26,186 to 24,931 under the baseline model but stabilizing at 30,943 under the adjusted model, preserving nearly 6,000 more individuals in a healthy weight range.

The overweight population initially grows before stabilizing. In the baseline scenario, it peaks and declines to 30,037 individuals by year 20, while under the adjusted model, it reaches 35,109, indicating interventions help prevent progression into obesity and the conversion of obesity to overweight. Obesity sees the most significant divergence, increasing from 41,897 to 43,524 in the baseline model, while the adjusted model stabilizes at 31,689 under the adjusted scenario, reducing obesity prevalence from 43.35% to 31.69%.

Overall, the adjusted scenario achieves a 26.90% relative reduction in obesity prevalence by helping obese policyholders to lose enough weight to be categorized as overweight and normal weight. These findings underscore the effectiveness of structured interventions in mitigating obesity trends, improving long-term health outcomes, and reducing obesity-related chronic disease risks.

We calculate the baseline insurance company expense in 2045 to be:

$$70\% \cdot (43,524 \cdot \$1,816 + 30,037 \cdot \$600) = \$67,943,248.80$$

The recommendation adjusted insurance company expense in 2045 are calculated:

$$70\% \cdot (31,689 \cdot \$1,816 + 35,109 \cdot \$600) = \$55,028,836.80$$

Our policy will save the insurance company \$12,914,412.

7 Conclusion

We used a machine learning approach to classify obesity risk and give obesity management recommendations using XGBoost and Monte Carlo simulations. By modifying lifestyle factors; namely physical activity, calorie monitoring, snacking, and hydration, obesity probability can be reduced by 36.14% and overweight can be reduced by 13.5%. A Markov Chain analysis over 20 years shows that implementing these interventions can lower obesity prevalence.

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